Chapter 35
Structured Products and Autocallables

This is where most people meet derivatives. Far more households own a structured note than have ever traded an option, and the note is sold by a bank branch or a private banker rather than by a broker.

Chapter 12 said structured products are packaged strategies and took two apart with their option prices assumed. Chapter 34 priced the components. This chapter puts them together and asks the question the previous two could not: what does the buyer actually own, and is the price fair? Section 35.10 answers with a number: a $1,000 note worth $979 the day it is sold, the $21 sitting inside the coupon rather than beside it.

The chapter is deliberately blunt about a conflict. Every instrument here is legitimate and most serve a real purpose. But the products are sold on a headline number (a participation rate, a coupon), and the risk sits in terms that do not appear in the headline. That gap is the subject.

Section 35.1 takes the common notes apart into a bond and options, and Sections 35.2 and 35.3 price the two that dominate, the reverse convertible and the autocallable, and show what the autocallable's return distribution looks like. Sections 35.4 and 35.5 cover the secondary market and the remaining families. Sections 35.6 to 35.8 turn to the issuer's side: how the embedded option is hedged, what happens when many notes share a barrier, and why most notes reference the worst of several underlyings. Sections 35.9 to 35.11 are about what the buyer pays for (the issuer's credit and the fee) and give a procedure for evaluating a note; Section 35.12 asks what disclosure can and cannot fix, and Section 35.13 who buys these products and why.

35.1 TAKING A NOTE APART

Every structured note decomposes into a bond and a derivative. Doing the decomposition is the single most useful skill in this chapter, because it converts a marketing document into a portfolio.

The decompositions are short, and the rest of the chapter refers back to them. Write F for the face amount, S_0 and S_T for the underlying at issue and at maturity, and $N = F/S_0$ for the number of option units a note of face F carries. In every case the redemption is a bond plus a position in options, and the only question is which options and on which side.

Principal-protected note. The buyer is promised the full face back at maturity whatever the underlying does, plus a share p, the participation rate, of any rise in the underlying over the note's life. A note with 90% participation on an index that rises 20% repays 118 per 100 of face; if the index falls, it repays 100. At maturity the redemption is

$$V_T = F + p\,N\,(S_T - S_0)^+ \tag{35.1}$$

a zero-coupon bond plus pN calls struck at the initial level. The bond secures the guarantee, and whatever the bond leaves over buys the calls, which is why the participation rate is a residual

rather than a promise, and why Example 35.1 finds it governed by the issuer's funding cost.

Capped participation. The same promise, except that the share of the rise is paid only up to a ceiling: above a level C the redemption stops growing. In exchange the participation below the cap is higher, often above 100%. Capping the upside at C turns the call into a call spread:

$$V_T = F + p\,N\left[(S_T - S_0)^+ - (S_T - C)^+\right] \tag{35.2}$$

The sold call is what pays for the higher headline participation. A note advertising a participation above 100% has almost always sold something, and this is usually what.

Reverse convertible. The buyer receives a fixed coupon c well above deposit rates. At maturity the face is repaid in cash if the underlying stock or index ends at or above a barrier H; if it ends below, the buyer receives shares, or their cash value, instead and so takes the whole fall from the initial level:

$$V_T = F(1 + c) - N\,(S_0 - S_T)^+\,\mathbf{1}_{\{S_T < H\}} \tag{35.3}$$

a bond and a coupon, less a put the investor has written: struck at the initial level, but paying only when the underlying ends below H. Setting $H = S_0$ deletes the indicator and leaves a plain short put, which is the un-barriered version of the product. *The indicator is also the discontinuity*: the note pays $F(1 + c)$ just above H and $F(1 + c) - N(S_0 - H)$ just below it, and Section 35.6 is about what that does to the hedge.

Shark fin. Principal is protected and the buyer shares in the rise, usually at a high participation rate, but only as long as the underlying never trades above an upper level H during the note's life. If it ever does, the participation is lost and the buyer receives a small fixed rebate R instead. The name comes from the shape of the payoff, drawn in the bottom-right panel of Figure 35.1: flat at par below the initial level, rising steeply along the participation line up to H, then dropping back to par plus the rebate, the outline of a shark's dorsal fin. At maturity

$$V_T = F + p\,N\,(S_T - S_0)^+\mathbf{1}_{\{\max_{t\le T} S_t < H\}} \;+\; R\,\mathbf{1}_{\{\max_{t\le T} S_t \ge H\}} \tag{35.4}$$

a bond, an up-and-out call, and the rebate. The knock-out is cheaper than the vanilla call of (35.1), so the same budget buys more of it: that, and not a better view, is where a shark fin's headline participation above 100% comes from. Shark fins come in two forms. Some observe the barrier only at maturity, so the payoff depends on S_T alone and Figure 35.1 draws it exactly; others observe it daily or continuously, which is what (35.4) writes and Example 35.4 prices, and then a path that touches H and falls back is knocked out too.

Autocallable. The buyer receives a high coupon, but the note can end early. On each of a series of observation dates, if the underlying is at or above a trigger level, the note is *called*: it repays the face plus all the coupons accrued so far, and stops. If it is never called, it repays the face with all the coupons at maturity unless the underlying ends below a barrier, in which case the buyer is repaid in shares, takes the loss as in a reverse convertible, and forfeits the coupons. In symbols, take observation dates $t_1 < \cdots < t_n = T$, an autocall trigger K_a, a coupon c per annum with each t_i measured in years, and a barrier H, observed here only at maturity for simplicity; in practice it is usually observed daily, as in Example 35.2. Let τ be the first date on which $S_{t_i} \ge K_a$, and $\tau = \infty$ if that never happens. Then

$$V = \sum_{i=1}^{n} F(1 + c\,t_i)\,\mathbf{1}_{\{\tau = i\}} \;+\; \left[F(1 + cT)\,\mathbf{1}_{\{S_T \ge H\}} + N S_T\,\mathbf{1}_{\{S_T < H\}}\right]\mathbf{1}_{\{\tau = \infty\}} \tag{35.5}$$

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a strip of digital payments, each conditional on not having been called already, plus, if the note is never called, the written put of (35.3): below the barrier the buyer receives shares worth NS_T and forfeits the coupons. Two things follow that the **term sheet** (the document setting out the note's terms for the buyer) does not say. *The investor is short volatility twice*: once in the digitals, which want the underlying quietly above K_a, and once in the put. And a reverse convertible is the case $n = 1$ with no trigger, except that it keeps its coupon below the barrier. The two share their downside, a written put that pays only below the barrier, and are sold to the same buyer on the same high coupon, but their risk profiles differ: early redemption cuts the autocallable's expected life well below its stated term, so the buyer collects the coupon for less time while carrying the same tail, and the outcome depends on the path as well as on where the underlying ends.

Dual-currency deposit. The buyer places a deposit in one currency and earns a rate above the ordinary deposit rate. At maturity the bank chooses the currency of repayment (the original one, or a second currency at a conversion rate fixed at the outset), and it will choose whichever is worth less to the depositor. Write X_T for the exchange rate at maturity, in units of the second currency per unit of the deposit currency, K for the conversion rate, and c for the interest, paid in the deposit currency whatever happens. Measured in the deposit currency, the redemption is

$$V_T = F(1 + c) - \frac{F}{X_T}\,(X_T - K)^+ \tag{35.6}$$

a deposit less a written call on the deposit currency struck at the conversion rate. When the deposit currency strengthens past K, the principal comes back converted at K and is worth only FK/X_T. Example 35.5 carries the arithmetic, and the enhancement is the option premium, as it is in every structure above.

One identity organizes all of them. At issue,

$$\text{issue price} = \underbrace{\text{PV(principal)} + \text{PV(coupons)}}_{\text{the bond}} + \underbrace{\text{options bought} - \text{options sold}}_{\text{the derivative}} + \text{fee} \tag{35.7}$$

Figure 35.1 draws the first four on one scale, and the shapes carry the argument. In each panel the dashed line is what the same 100 would return left on deposit, at 4% for a year, so the gap between it and the solid line is what the embedded options add or take away against simply earning interest. The principal-protected note repays 100 where the deposit pays 104, which is the interest given up to buy the participation; the reverse convertible pays 110 above its barrier, the extra 6 being the premium for the put the buyer has written, and falls far below the deposit once the barrier is breached. Where each solid line crosses the dashed one marks the range of the underlying in which the buyer comes out ahead:

- above 104.4 for the principal-protected note;
- above 104 for the capped note;
- anywhere at or above the barrier of 70 for the reverse convertible;
- only between 103.6 and the knock-out at 130 for the shark fin.

Each note is a bet that the underlying ends in its own range, and the reverse convertible's range is the widest because its losses, when they come, are the largest. Every term but the fee is quotable, and that is what makes the decomposition worth doing: price the bond at the issuer's own funding curve, price the options in the market, and the residual is what the note costs to own.

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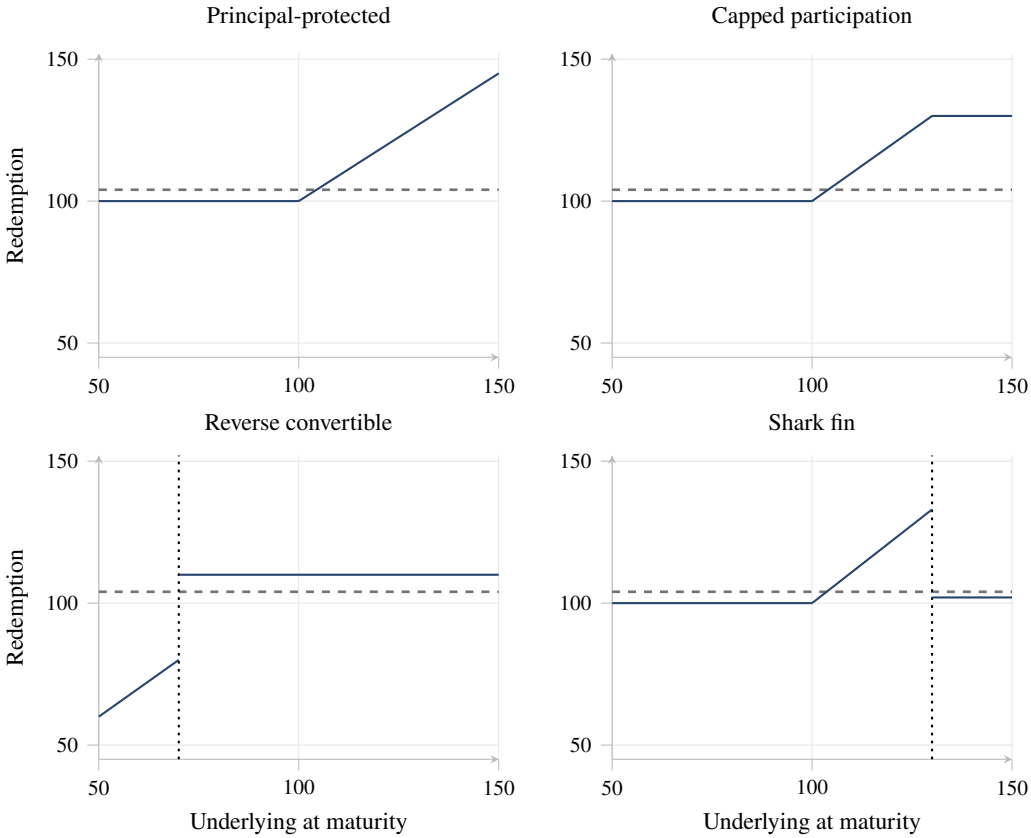


Figure 35.1: Redemption per 100 of face, initial level 100. Participation 90%, cap 130, barrier 70 with a 10% coupon, knock-out 130 with a 2% rebate. Dashed: 100 left on deposit at 4% for a year. Illustrative.

Example 35.1 Where the participation rate comes from

Example 12.7 took this note apart: five years, a zero-coupon bond worth \$79.85 per \$100 of face at 4.5% (\$798.52 on the \$1,000 face used here), the option budget it leaves, and what a 2% fee takes out of that budget. It held the call's price fixed. Here the index yields 2.0% with 20% volatility, and the issuer's own funding rate discounts the at-the-money call as well as the bond (the call costs \$209.60 at 4.5%), so both legs move with funding:

Issuer funding rate	Participation (no fee)	(2% fee)
2.0%	59.4%	46.9%
3.0%	77.7%	66.6%
4.5%	96.1%	86.6%
6.0%	107.3%	99.0%

The participation rate is mostly a statement about the issuer's funding cost, not about the index. It nearly doubles as funding goes from 2% to 6%, with the index unchanged.

So across issuers at one moment, a higher participation rate signals a weaker credit. The issuer paying more to fund can buy more optionality, the best-looking note in a comparison is systematically the one with the most credit risk, and Section 35.9 says what that has cost.

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35.2 REVERSE CONVERTIBLES AND AUTOCALLABLES

The reverse convertible comes first, because an autocallable is a reverse convertible with early redemption added. The investor in a reverse convertible is short the put of (35.3), and the coupon is the premium, as Example 12.8 showed: a note paying 10% when deposits pay 4% is not offering 6% of extra yield; it is paying 6% for a put the investor has written. The mis-selling risk is structural rather than incidental. The product is described in the vocabulary of fixed income (coupon, maturity, principal), and its risk is entirely equity, so a buyer who compares it with a bond is comparing the wrong thing.

The autocallable is the dominant structured product globally, and the rest of this chapter is built on it.

The mechanics are simple to state and hard to price. The note pays a coupon and terminates early ("autocalls") if the underlying is above a level on an observation date. If it never autocalls, principal is returned at maturity together with the accrued coupons *unless* the underlying has breached a lower knock-in barrier, in which case the investor takes the loss and forfeits the coupons.

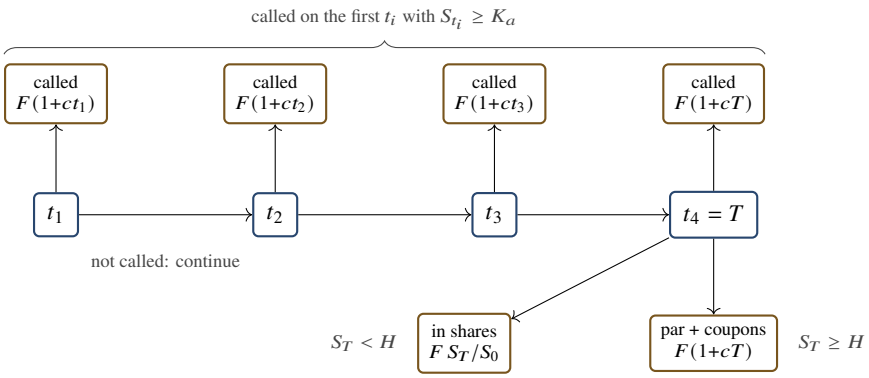


Figure 35.2: Each observation either redeems the note with the coupons accumulated to that point or passes it on. Only the final date can end below the barrier.

Figure 35.2 is (35.5) drawn. The note is a sequence of chances to be redeemed early, each paying the coupons banked so far, and only the last one can end in shares.

Example 35.2 Pricing one, and what the buyer owns

A two-year note on an index at 100. Autocall at 100% observed monthly after a two-month **lockout period**, during which the note cannot be called, so the first observation is at month three; knock-in at 75% observed daily. The coupon accrues at an annual rate c: a note called at the end of month m repays $1 + c\,m/12$, and one that is never called and never knocked in repays $1 + 2c$ at maturity. If knocked in and never autocalled, the investor receives $\min(S_T/100, 1)$ per unit of face and no coupon. Volatility 25%, rate 2.5%; 80 million antithetic paths (200 seeds of 400,000), which put the fair coupon's standard error at 0.005 of a percentage point. The coupon c is the unknown: it is set so that the note is worth exactly its issue price of 1.00, which makes it the *fair coupon*.

P(autocalls at some point)	0.8311
P(knocks in at some point)	0.4132
P(**knocked in and never autocalled**)	**0.1645**
Fair coupon	**18.42% per annum**

An 18.4% coupon, and the note prices at par. Nothing is being given away: the coupon is

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exactly what the embedded short put is worth.

The probability structure is what sells it and what should worry the buyer. The note produces its advertised return about 83% of the time. That is why these products are easy to sell and why holders report satisfaction for years at a stretch.

The remaining 17% is where all the loss is, and it is not a small loss—a knocked-in note pays $\min(S_T/100, 1)$, so the investor absorbs the full downside from par.

This is a negatively skewed payoff (frequent small gains, rare large losses) sold on its mean and its hit rate, which is the same structure as Section 22.8's inverse volatility products and Section 33.6's range accrual, and it recurs because it is what sells.

Now consider the exposure the buyer does not see:

Realized volatility	**Note value at** $c = 18.42\%$
15%	1.0520
20%	1.0204
25%	**1.0000**
30%	0.9847
40%	0.9574

The note is short volatility. It was priced at 25%; realized volatility above that makes it worth less than par, and the investor never agreed to a volatility view.

And it is short volatility in the worst way, because volatility rises when markets fall, which is exactly when the knock-in is approached. The two risks are not independent—this is Section 19.9's leverage effect and Section 27.9's wrong-way structure appearing in a retail product.

35.3 WHAT THE RETURN DISTRIBUTION LOOKS LIKE

A fair price says the mean is right. It says nothing about the shape, and the shape is what the buyer is actually choosing.

Example 35.3 The same expected return, a different distribution

Simulate Example 35.2's note at its fair 18.42% coupon on the same 80 million paths, and record the investor's total return, then do the same for simply holding the index for two years.

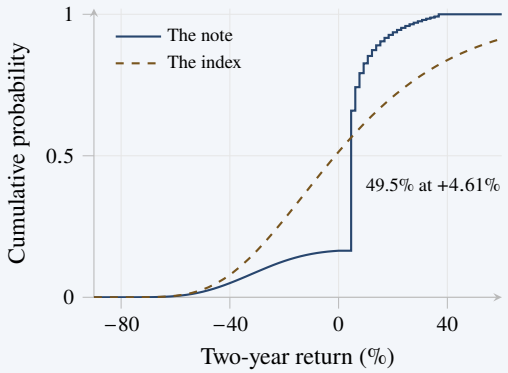
	The note	**The index**
Mean total return	+1.60%	+5.13%
Standard deviation	17.25%	38.36%
Skewness	**−1.59**	**+1.14**
Excess kurtosis	2.83	—
P(positive)	**83.55%**	48.59%
P(loss > 25%)	11.02%	—
5th percentile	−40.03%	−44.79%

Everything here is under the pricing measure, so both instruments earn the risk-free rate by construction. The note's lower mean is not a worse deal: it terminates early, after 0.70 years on average, so it earns that rate over a shorter life. $2.5\% \times 0.70 = 1.75\%$, against the simulated 1.60%, which falls a little short because the paths that run longest, and so carry the largest growth factor, are the knocked-in ones that lose; that is the internal check that the pricing is consistent.

With the mean neutralized, the comparison is purely about shape, and the shape is the product.

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Median return	+4.61%
E[return | positive]	+8.22%
E[return | negative]	**−32.05**%
Average of the worst 1%	−60.23%



Wins average 8% and losses average 32%—a four-to-one asymmetry, at a five-to-one win rate. Those two ratios are the product, and they multiply out to a fair price.

Three observations follow.

Skewness measures a distribution's asymmetry: it is negative when the long tail lies on the loss side, so that small gains are frequent and large losses rare, and positive when the long tail lies on the gain side. (Excess kurtosis, the row below it, measures how much more of the distribution sits in the tails than a normal distribution's, for which it is zero.) The note's skewness is negative and the index's is positive. A buyer who moves from the index into the note has not reduced risk; they have exchanged a positively skewed exposure for a negatively skewed one at the same expected return, which is the opposite of what "capital-efficient" and "defensive" suggest.

The 25th and 50th percentiles are identical, at +4.61%. Nearly half the distribution, 49.5%, is a single point (the note autocalled at the first opportunity), so every percentile from the 17th to the 65th is +4.61%, which is why holders experience these products as reliable. The plot beside the second table draws both distributions cumulatively: the note's single point is the 49.5-point vertical step at +4.61%. Below zero the note's curve lies under the index's, because a loss is far less likely (16.45% against 51.41%), but the note can never pay more than +36.84%, the level by which the index has reached only its 82nd percentile.

And the standard deviation is genuinely lower than the index's, 17.2% against 38.4%. A risk measure that sees only the second moment will report this product as safer, which it is on that measure and is not on any measure that looks at the tail. Chapter 38's warning that a single-number risk measure can miss what sits in the tail is not abstract; *this is the product it describes*.

35.4 THE SECONDARY MARKET

A structured note is quoted daily and is not liquid, and the distinction matters more than it sounds.

The issuer is usually the only bid. There is no exchange, no competing market maker, and no obligation to quote. A buyer who wants out sells back to the party who sold it to them, at a price that party computes.

Three consequences follow.

The bid embeds a second spread. Having paid the issue premium of Section 35.10, an early seller pays again on the way out. Round-tripping a note in the first year can cost several percent of principal before any market move.

The quoted price is a model output. These are Level 2 or Level 3 valuations in the accounting sense (prices computed from a model, with observable or unobservable inputs respectively, because no active market quotes the instrument), and Section 26.12's observation about the crisis applies in miniature: when the only price comes from the party on the other side, the holder cannot distinguish a mark from a quote.

And illiquidity interacts with the payoff. An investor most wants to exit after a sharp fall,

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when the knock-in is close and the outlook has changed, and that is exactly when the bid is lowest and the embedded option's value has moved most against them. The exit is worst when it is most wanted, which is this book's recurring wrong-way structure appearing one last time.

Case Study 35.1 A market that learned this from its losses

South Korea developed one of the world's largest retail markets for autocallable notes, sold as **equity-linked securities** through bank branches to ordinary savers. A large share referenced the Hang Seng China Enterprises Index (HSCEI), chosen because its high volatility supported high coupons.

The structure worked as designed for years. Notes autocalled, holders were repaid early with the advertised coupon, and the proceeds were rolled into new notes: the 83% outcome of Example 35.3, repeated often enough to look like a rule rather than a probability.

Then the index fell a long way and stayed there. Notes issued in 2021 carried knock-in barriers set against 2021 levels; by the time those notes reached maturity in 2024, the index had not recovered, and the loss that Example 35.2 gives a 17% probability became the realized outcome for a large cohort at once.

Four features of the episode are general and worth separating from the specifics.

The losses were synchronized because the issuance was. Notes sold during one period share a vintage, a barrier level, and a maturity. A product with a 17% loss probability sold to a hundred thousand people in the same quarter does not produce 17% of them losing independently; it produces all or none.

The distribution channel mattered as much as the product. The notes were sold through bank branches to savers, alongside deposits, by staff whose training and incentives were not designed for a negatively skewed derivative.

The regulatory response focused on sales practice rather than on the instrument: **suitability** assessment (a check, before the sale, that a product fits the buyer's circumstances, knowledge and objectives), recording requirements, limits on where such products may be sold. That is the right diagnosis if one accepts that the instrument is legitimate and the mismatch was between the payoff and the buyer.

And the mechanism was entirely visible in advance. Everything in Sections 35.2 through 35.3 could have been computed at issue from the term sheet. Nothing about this episode required hindsight, which is what distinguishes it from most of the case studies in this book.

35.5 OTHER FAMILIES

Four more structures account for most of the rest of the market, and each is a recombination of what has already been priced.

Shark-fin notes pay an enhanced participation up to a barrier and a small fixed rebate if the barrier is touched: an up-and-out call plus a rebate. *The buyer is selling the tail of the upside*, and the appeal is that giving up the potential of returns above 30% feels costless to someone who expects 10%. Example 35.4 prices what the feeling costs.

Example 35.4 Where the headline participation comes from

A one-year \$1,000 note at a 3% funding rate, on an index at 100 with 25% volatility. The zero-coupon bond costs 970.45, leaving **29.55** for the option and its rebate.

An at-the-money call is worth 11.3485 per 100 of index, so that budget buys **26.0%** participation, an unattractive number to print on a term sheet. Adding a knock-out at 130 drops the option

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to 2.1766, but a shark fin also pays a rebate, here 2% of face at maturity if the barrier is touched. The touch probability, from Section 34.3's formula, is 0.292, so the rebate costs $1{,}000 \times 2\% \times e^{-0.03} \times 0.292 = 5.68$, and the remaining 23.88 buys

$$\frac{23.88}{21.766} = \textbf{109.7\% participation}$$

The participation rate has more than quadrupled, and what has been given up is every path that touches 130 at any time before maturity, not only those that end above it.

The question the term sheet does not answer is how often those outcomes occur. That touch probability, **29.2%** under the pricing measure, is the answer: in nearly a third of scenarios the 109.7% participation is extinguished and the note pays its small rebate—and the knock-out happens in the states where the index rallied hardest. The buyer has not sold a remote tail; they have sold roughly the three strongest rallies in ten.

Two features make this harder to see than the autocallable's. When the barrier is monitored continuously, as here, a single intraday touch during a rally destroys the participation for the rest of the year even if the index closes the year at 128. And the rebate is quoted as a percentage of face rather than as what it replaces: a 2% rebate against a forgone 109.7% participation on a 30% move is 2% against 32.9%. The instrument is Example 34.6's reverse knock-out sold retail, with the same discontinuity and none of the warning.

Dual-currency deposits pay an above-market rate in one currency, with the issuer holding the right to repay in another at a preset rate. *It is a deposit plus a short FX option*, and it is among the most widely sold structured products in Asian private banking. The presentation as a "deposit" is the problem: the principal is at risk in currency terms and the vocabulary says otherwise. Example 35.5 shows that the entire enhancement is the option premium, to the basis point.

Example 35.5 The deposit that is not one

A three-month dollar deposit of $100,000 with a conversion rate of 153 against a USD/JPY spot of 150. Dollar deposits pay 4.5%, yen 0.5%, and three-month volatility is 10%. The depositor has written a dollar call struck at 153: if the dollar strengthens past it, the bank repays in yen at 153. Interest is paid in dollars whatever happens; only the principal can be converted.

Chapter 16's Black–Scholes–Merton formula, with the yen rate for discounting and the dollar rate as the yield q (the form known as Garman–Kohlhagen), values that option at **1.2776** yen per dollar, which is 0.852% of the dollar notional over the quarter, or **3.41% annualized**. So:

$$4.5\%\ \text{(deposit)} \ + \ 3.41\%\ \text{(written option)} \ = \ \textbf{7.91\%}$$

The advertised 7.9% rate is the deposit rate plus the premium, and nothing else. A depositor who understood the decomposition would ask whether 3.41% a year is a fair price for the option—which is the question the product's presentation prevents, because a deposit rate is not something one negotiates.

The outcomes are asymmetric in the familiar direction:

USD/JPY at maturity	Repaid in	Quarter's return
140	dollars	+1.98%
150	dollars	+1.98%
153	dollars	+1.98%
160	yen at 153	−2.40%
165	yen at 153	−5.30%
175	yen at 153	−10.59%

The gain is capped at one quarter's coupon and the loss is not capped at all, which is Section 35.2's shape in a three-month product and a different asset class. And the conversion

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arrives at the worst moment: a dollar-based investor is repaid in yen precisely when the yen is weak.

Credit-linked notes embed the short CDS position of Section 25.14. The whole of Part V's credit machinery applies, and the retail versions are the ones where the reference entity's identity is least understood by the buyer.

And range accruals appear as notes as well as swaps, with the coupon arithmetic and the short-vega exposure of Section 33.6 unchanged.

Hybrids attach a second condition from another asset class to the credit one (a coupon paid only while an issuer survives *and* an index is above its strike), and they are the exception to this section's arithmetic. The two legs cannot be priced separately and added, because the value depends on a correlation between default and the other factor that neither market quotes; Section 25.15 works the example, and the practical consequence for a buyer is that the usual decomposition is unavailable to them as well as to the dealer.

The pattern across the first four is the one this chapter began with. The enhanced headline term is paid for by an option the buyer has written, and the option is described by its trigger rather than by its risk.

The exchange-traded buffer fund is the contrast case. A defined-outcome fund holds a deep in-the-money call, a put spread that supplies the buffer, and a sold call that finances the cap, and it resets each year. Its cap and buffer are quoted, its terms are rules rather than a negotiated term sheet, no issuer's credit is involved, and its cost shows in the expense ratio, so it supplies the comparison for Sections 35.9 and 35.10.

35.6 ISSUER HEDGING, AND THE DISCONTINUITY

The investor's position implies the issuer's, and the issuer's hedging is where these products stop being a private matter between two parties.

The investor is short a down-and-in put; the issuer is long it. A long put has negative delta, so the issuer hedges by holding the index, and the size of that holding is what matters.

Example 35.6 Why the hedge stabilizes, until it does not

The embedded option: a down-and-in put struck at 100 with a barrier at 75, one year, with Example 35.2's 25% volatility and 2.5% rate and no dividends. Valued exactly with the down-and-in put row of Section 34.3's barrier formulas. The note's knock-in is observed daily, which Section 34.6's correction handles by moving the barrier away from the spot, to 74.32; that lowers the value at 100 from 6.160 to 5.956 and leaves the shape of the delta, and its jump at the barrier, essentially unchanged, so the continuous barrier is used below.

Index	Option value	Delta
130	0.706	−0.055
110	3.128	−0.220
100	6.160	−0.399
90	11.365	−0.652
80	19.262	−0.921
76	23.124	−1.006
75.33	23.802	**−1.017**
74.67	24.411	**−0.827**
74	24.968	−0.836
70	28.414	−0.885

Above the barrier, delta steepens monotonically from −0.055 to −1.017 as the market falls, and it *exceeds one in magnitude*, a barrier feature Section 34.3 flagged.

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The issuer, long this option, must therefore hold a growing long position in the index as the market falls. From 130 down to the barrier the hedge grows from 5.5% to 101.7% of notional. The issuer is buying into a falling market, continuously, and that flow is stabilizing. It is the gamma term of Chapter 18's expansion, equation (18.1), with the sign Section 18.12 showed dampens a move: above the barrier the issuer is long gamma, so keeping its delta term canceled means buying as the market falls.

Then the barrier is touched and the delta jumps from −1.017 to −0.827: a change of +0.19.

Two things happen at once and the second is the larger. The issuer must *sell* about 19% of notional immediately. And *the buying stops*: below the barrier the option is an ordinary put whose delta steepens far more slowly, so the persistent bid that had been supporting the market on the way down simply disappears.

The popular account of these episodes emphasizes the forced selling. The arithmetic says the withdrawal of the bid is the bigger effect, and stating it that way changes what a regulator should monitor: not the size of the sell order at the barrier, but the size of the stabilizing flow that is about to vanish.

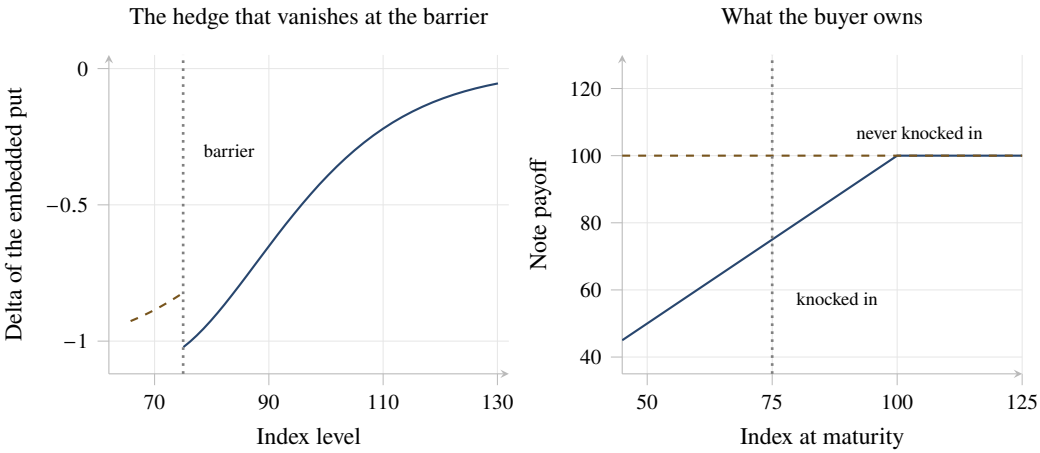


Figure 35.3: Left: Example 35.6's delta. Right: maturity payoff, coupon excluded.

Figure 35.3's left panel is the mechanism in one picture. The gap at the barrier is the discontinuity, and the steep approach to it on the left is the stabilizing flow that ends there. Its right panel is the maturity payoff, coupon excluded: flat at par if the barrier was never touched, and the index level itself, up to par, if it was.

35.7 WHEN MANY NOTES SHARE A BARRIER

One note's hedging is irrelevant to a market. The problem is that barriers cluster.

Notes are issued continuously against a rolling reference level, so a sustained decline of a given size brings a large cohort of barriers into range at once. The clustering is not coincidence but arithmetic: barriers are set at round fractions (75%, 80%) of the level prevailing at issue, so notes issued during any period of stable prices share nearly the same absolute barrier.

Three consequences follow, in order of increasing seriousness.

The stabilizing flow is larger than any one issuer knows. Each dealer sees its own book, which is Section 27.13's aggregation problem in a different market.

The withdrawal is synchronized. When the index passes the cluster, many issuers' bids disappear within the same few sessions.

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And the feedback is self-reinforcing while it lasts. The buying on the way down slows the decline, which keeps the index above the barriers longer, which sustains the buying—until it does not. A stabilizer that works until it stops is not a stabilizer, and Section 23.10's saturating pin risk made the identical point about a different product.

Case Study 35.2 A cluster passed through: China, January–February 2024

China's structured-product market is built around the **snowball**, the domestic form of the autocallable, and its standard terms track Example 35.2 closely: a CSI 500 or CSI 1000 underlying (the China Securities Index Company's mid-cap and small-cap indices), twelve to twenty-four months, monthly knock-out (autocall) observation after a lockout of about three months, a knock-in observed daily at 70–80%, and coupons of 15–20% a year. The underlying is chosen for its volatility, which is what pays the coupon (the same reason Korea's notes referenced an offshore China index), and that coupon range is the same order as the 18.42% of Example 35.2, because it is the same option at the same volatility.

One feature of the market differs from Korea's in the way that matters most. Snowballs are sold by securities firms acting as counterparty, wrapped in asset-management products with a minimum subscription around RMB 1 million, so the buyer is a qualified investor rather than a depositor. The second of the Korean episode's four general features, that the distribution channel matters as much as the payoff, had been addressed institutionally, and in advance.

What had not been addressed was the synchronization. Barriers were set at 75% or 80% of the index level prevailing at issue, so notes written through a comparatively stable 2023 shared nearly the same absolute knock-in level. The decline of January and February 2024 went straight through it.

Both of Example 35.6's effects are identifiable in what followed, and the popular account has their sizes the wrong way round. Dealers hedging the long down-and-in put had been buying index futures as the market fell, and that flow was supporting it. At the barrier the immediate sale is about 19% of notional, while the larger effect is that the bid supporting the market on the way down disappeared. So snowball hedging did not cause the fall; it was a mechanism that first slowed the decline and then stopped slowing it—and in a price series, "stopped slowing" looks a great deal like "caused."

Attribution is harder still because a second leveraged chain, quantitative funds' equity positions financed through broker swaps, was unwinding in the same week, and separating the two needs dealer-level position data that is not public.

The regulatory response is the informative part, because most of it ran ahead of the event. Suitability requirements for snowball sales, an explicit instruction that the product is not principal-protected, and later limits on how much of a private fund could be put into snowballs (reported from 2022) and on how much a dealer could issue against its capital were all in place before 2024. *None of them addressed the clustering*, because suitability governs who may buy, and the cluster is a fact about how many investors bought notes with nearly the same barrier level in the same quarter. A rule aimed at individual prudence does not solve a problem about an aggregate, which is Section 27.13's conclusion, reached again in a different market. The measures after the cluster confirmed the diagnosis. Press reports in April 2024 said regulators had told some large brokers to halt any rise in their net exposure to over-the-counter derivatives linked to A shares, snowballs included, and on October 25, 2024 the Securities Association of China issued measures on income certificates under which a floating-income certificate's loss on its underlying may not exceed its fixed-income amount.

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35.8 WORST-OF UNDERLYINGS

Section 34.11 priced the mechanism; here is what it does commercially.

Most autocallables reference the *worst* of two to four underlyings. *This roughly doubles the coupon*, and Example 34.9 showed why: at a correlation of 0.20 a worst-of put on three assets is worth 2.05 times a single-asset put.

The disclosure problem is acute and simple. A note "linked to three major indices" sounds diversified. It is the opposite: the investor is exposed to whichever of the three performs worst, so adding an underlying can only increase the risk and can only increase the coupon.

And the coupon's size is the tell. A coupon far above what a single-underlying note pays is not a better deal; it is a measure of how much extra risk was added, and it is the one number in the term sheet that reliably signals the risk it was designed to obscure.

35.9 THE ISSUER'S CREDIT

A structured note is an *unsecured obligation of the issuer*. The derivative is embedded in a bond, and if the bond defaults the derivative goes with it.

Principal protection means protection from the market, not from the issuer. That distinction was academic until September 2008, when retail holders of notes issued by a failed investment bank discovered that "100% principal protected" ranked alongside every other senior unsecured claim.

The pricing consequence is Example 35.1's uncomfortable corollary. The issuer's funding spread *is* the source of the note's attractive terms, so, other terms equal, a better-looking note is a worse credit. The investor is being paid for taking issuer credit risk, and in almost no case is that presented as the reason the terms are good.

Two developments have improved this and neither removes it. Disclosure rules now require the issuer's credit to be stated, and some jurisdictions require an estimated initial value (the note's fair value net of fees) to be printed on the term sheet. The estimated-value disclosure is the single most effective reform in this chapter's subject, because it makes Example 35.7's fee arithmetic visible without requiring the buyer to do it.

35.10 WHERE THE FEES ARE

The economics of the business are worth stating plainly, because they explain the product design.

A structured note carries no explicit management fee. The compensation is the difference between the note's issue price and its fair value, typically 1.5% to 3% of face for a simple structure and more for a complex one, earned at issue.

This creates a specific incentive: complexity is profitable because it is hard to price. A buyer can check a participation rate against a competitor's; the fair value of a worst-of autocallable on three underlyings with a monthly autocall and a daily knock-in is not something a buyer can check at all. The margin is largest where the price is hardest to verify, which is the same structure Section 33.13 identified in non-standard swaps and Section 34.13 in exotics.

The fee is not an estimate once the fair coupon is known, and Example 35.7 extracts it from the numbers already computed.

Example 35.7 Reading the fee out of the coupon

Example 35.2's note prices at par on a coupon of **18.42%**. Suppose the same structure is issued at 1,000 paying **12.65%**.

The factor that converts a coupon into value is not the 0.70-year average life of Section 35.3. A note that knocks in and is never called runs the full two years and pays no coupon at all, and those

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paths carry nearly half of that average; what remains is then discounted. Simulating the note gives the factor directly: each point of annual coupon is worth **0.366** points of face. So the value given up is

$$(18.42\% - 12.65\%) \times 0.366 = 2.11\% \text{ of face}$$

and the note's fair value at issue is about **979** against an issue price of 1,000. The $21 is the fee, and it sits inside the coupon rather than beside it.

Three things follow that a term sheet does not say. The fee is paid at issue and cannot be recovered by holding to maturity: it is not an annual charge of 2.11% but a single 2.11% taken on day one, which on a note with a 0.70-year expected life is an annualized **3.0%**. *The buyer who sells early pays it twice*, once at issue and again in Section 35.4's exit spread. And *the fee is invisible in every scenario table*, because those tables are computed from the stated coupon and the stated coupon is already net of it.

This is precisely what the estimated-initial-value disclosure repairs. Printing "estimated initial value: $979 per $1,000" converts an argument about fairness into a number a buyer can compare across issuers, without requiring anyone to understand a single thing about barrier options. It does not attempt to explain the product; it prices it.

Illustrative magnitudes follow, scaled to complexity:

	Fair value	Gap to par
Principal-protected note, one index	0.9800	2.00%
Reverse convertible, one index	0.9750	2.50%
Autocallable, one index	0.9700	3.00%
Autocallable, worst-of three	0.9550	**4.50%**

The gap widens exactly as the buyer's ability to check it narrows.

The defense, which deserves stating, is that the alternative is worse for the buyer. These structures cannot be assembled retail: the components are OTC, the sizes are institutional, and the hedging is continuous. A margin for manufacturing a payoff that could not otherwise be bought is a legitimate charge. The question is its size and whether it is visible, not whether it should exist.

35.11 HOW TO EVALUATE A NOTE

The chapter's material assembles into a procedure. It is short, it requires no model the reader does not now have, and it answers the questions a term sheet is designed not to raise. It has five steps:

1. *Decompose it.* Every note is a bond plus one or more options. Identify the bond, identify each option, and, critically, identify which side of each option the buyer is on. In Example 35.1 the buyer is long a call; in Section 35.2's reverse convertible and Example 35.2 the buyer is short a put. If the headline term is attractive, the buyer is short something.

2. *Price the bond leg at the issuer's own curve.* Example 35.1 showed that an attractive participation rate comes mostly from the issuer's funding spread: an issuer that pays more to borrow spends less on the zero-coupon bond and has more left to buy options. A note from an AA issuer and one from a BBB issuer are not comparable on their headline terms, and the difference between them is credit compensation, not investment skill.

3. *Value the option leg and compare with par.* The gap is the fee. Where an estimated initial value is disclosed, this step is done for the reader; where it is not, the table in Section 35.10 gives the order of magnitude, and the rule that it widens with complexity is reliable.

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4. *Simulate the distribution, not the scenarios.* A term sheet presents three or four illustrative outcomes. Example 35.3 shows what the term sheet omits: the skewness, the conditional loss given a loss, and the fact that the median and the 25th percentile can be the same number. A buyer who sees only $E[\text{return}]$ and $P(\text{positive})$ has seen the two statistics the product was engineered to optimize.

5. *Ask what the exit is worth.* Section 35.4: the issuer is the only bid, the spread applies again, and the bid is lowest when the need to sell is greatest. A note should be evaluated as a hold-to-maturity instrument, because that is what it is.

Applied to Example 35.2's note, the five steps produce a description no marketing document contains: a two-year unsecured loan to the issuer, bundled with a written down-and-in put on an index, sold, at Example 35.7's 12.65% coupon, about 2.1% above fair value, returning a positive amount with 83% probability (most often 3.2%, on average 5.6%) and on average −32% with 17% probability, and not sellable except back to the issuer.

That description is neither hostile nor favorable. It is the instrument, and a buyer who wants it after reading it is making an informed choice—which is the entire objective.

35.12 SUITABILITY, AND WHAT DISCLOSURE CANNOT FIX

Disclosure is necessary and it is not sufficient, and this chapter's material shows precisely where it stops working.

What disclosure fixes well. Fees and fair value: the estimated-initial-value requirement converts an invisible cost into a printed number, and it works because the quantity is a single figure the issuer already computes. Section 35.10's gap is exactly the kind of thing disclosure is good at.

What it fixes poorly. A distribution cannot be disclosed into intuition. A buyer told that the product has a skewness of −1.59 has been given a true fact and no understanding, and a buyer shown three illustrative scenarios has been given an understanding that is wrong. The gap between Example 35.3's two columns is not a disclosure failure; it is a comprehension problem, and jurisdictions that have tried to solve it with more disclosure have generally produced longer documents and unchanged outcomes.

What it cannot fix at all is the correlation between the product's failure and everything else the buyer owns. An autocallable knocks in when equity markets fall, which is when the buyer's other assets have fallen, their employer is under pressure, and their appetite for risk has collapsed. No disclosure of the product in isolation reveals that, because the fact is about the portfolio and not about the note.

That points at the reform this chapter would actually endorse: not more disclosure, but a suitability test framed around the payoff's shape rather than the buyer's stated risk tolerance, because a buyer who says they want "income with capital preservation" has described a payoff the product does not have, and no amount of the buyer confirming that they understand the risks changes that mismatch.

Two limits on that recommendation need stating, so that the argument is fair. Suitability regimes are paternalistic and impose real costs, including on buyers who understand the product perfectly well. And the demand is genuine: the yield problem these products solve is real, and a regime that removes them without addressing that demand pushes it somewhere less regulated. This is a trade-off rather than a solved problem, which is why Section 35.13 names the buyers rather than dismissing them.

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35.13 WHO BUYS THESE, AND WHY

Retail and private-bank clients are the volume, and the demand is genuine rather than manufactured. The dominant motive is yield in a world where deposits pay less than the buyer needs, and the second is a preference for defined outcomes over open-ended ones. Both are real preferences and neither is irrational.

Insurers and pension funds buy principal-protected structures for the reason Section 31.17 gave: their liabilities contain guarantees, and a note with a floor matches them better than an unhedged equity holding.

Corporates occasionally buy them as treasury investments, and the suitability question is sharper there than in retail, because the sums are larger and the governance is often thinner.

Issuers sell them for funding and for the margin. The funding motive is underrated: a structured note is term funding raised from a source that does not otherwise lend to banks, and it is stickier than wholesale funding because the buyer cannot easily exit.

And dealers, frequently a different institution from the issuer, take the hedging risk, which is where Section 35.6's flows originate.

The recurring conclusion of the preceding three chapters applies here in its sharpest form. The client is short the optionality and the dealer is long it, and here the client is a household. That does not make the trade illegitimate, but it does mean the disclosure question is not a formality—it is the whole of the consumer-protection issue, and Section 35.9's estimated-value requirement is the reform that addresses it most directly.

SUMMARY

Every structured note decomposes into a bond and a derivative, and doing the decomposition converts a marketing document into a portfolio.

A principal-protected note spends most of its face on the zero-coupon bond and buys participation with the rest, and the distributor's fee cuts that participation noticeably. Participation rises steeply with the issuer's funding rate while the index is unchanged, so the best-looking note on offer is systematically the weakest credit. A reverse convertible is a bond plus a short put that pays only below its barrier: its coupon above deposit rates is payment for a written put, described in fixed-income vocabulary while its risk is entirely equity.

An autocallable with a daily knock-in prices at par on a coupon far above deposit rates. It pays its advertised return most of the time and loses in the rest, with small frequent gains set against rare large losses, so its returns are negatively skewed where the index's are positively skewed. A buyer moving from the index into the note exchanges a positively skewed exposure for a negatively skewed one at the same expected return, and a variance-based risk measure reports the note as the safer of the two. It is also short volatility at the worst time, since volatility rises as the knock-in is approached.

Korea's retail market showed what this does at scale: the usual outcome repeated until it looked like a rule, and then a single vintage's barriers, maturities, and losses arrived together. A loss probability sold to a hundred thousand people in one quarter does not produce a matching fraction of them losing independently; it produces all or none.

The issuer is long the embedded down-and-in put, whose delta steepens sharply just above the barrier and jumps at it. The withdrawal of the issuer's bid matters more than the forced sale, and because barriers cluster at round fractions of the level at issue, the withdrawal is synchronized across issuers who each see only their own book. Worst-of underlyings raise the coupon substantially, and a note "linked to three major indices" is the opposite of diversified.

A note is unsecured issuer debt, so principal protection is protection from the market, not from the issuer. The fee is largest where the price is hardest to verify; disclosure fixes fees well and distributions poorly. The secondary market is the issuer alone, so the exit is worst when it

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is most wanted. The reform this chapter would endorse is a suitability test framed around the payoff's shape rather than the buyer's stated risk tolerance.

FURTHER READING

The regulatory literature is where this subject is treated, because supervisors have had to define what a fair disclosure looks like: the estimated-value rules and the key information document of the European Union's PRIIPs Regulation (No. 1286/2014) are worth reading as primary sources, and their differences across jurisdictions are informative about what each regulator thought the problem was. On the academic side, the work measuring issue prices against model fair values (Henderson and Pearson; Célérier and Vallée) is consistent enough across markets and decades to be treated as settled: notes are issued at a premium to fair value, and the premium rises with complexity. Bergstresser and others on the retail structured-products market document the demand side. On autocallable hedging flows the public literature is thin and the useful material is in central-bank financial stability reports, which have covered the topic increasingly since 2022. For the pricing machinery, the references of Chapter 34 apply unchanged, since an autocallable is a barrier product with an early-termination feature and nothing in it is new.

PRACTICE QUESTIONS

35.1 Using Example 35.1, compute the participation rate before and after a 2% fee, and explain why the same note's participation would be 59.4% at 2% issuer funding and 107.3% at 6%.

35.2 Explain why the best-looking note in a comparison across issuers at one moment is systematically the weakest credit.

35.3 Decompose a reverse convertible paying 10% when deposits pay 4%, and state what the 6% actually buys.

35.4 Using Example 35.2, state the fair coupon and the three probabilities, and explain why a payoff sold on its hit rate is negatively skewed.

35.5 Explain why an autocallable is short volatility in the worst possible way, being specific about when volatility rises.

35.6 State the note's and the index's skewness from the simulated distribution, and explain what a buyer switching between them is exchanging when both earn the risk-free rate.

35.7 State the average win, the average loss, and the worst one percent, and reconcile them with the 83.55% probability of a positive return.

35.8 Explain why the 25th and 50th percentiles of the note's return coincide, and why that makes the product unusually persuasive.

35.9 Explain why a variance-based risk measure reports this note as safer than the index, and what that says about variance as a risk measure.

35.10 Explain why the exit from a structured note is worst precisely when it is most wanted.

35.11 Explain why a product with a 17% loss probability sold to a hundred thousand people does not produce 17% of them losing independently.

35.12 Using Example 35.6, state the delta just above and just below the barrier and the jump, then explain why the withdrawal of the issuer's bid matters more than the forced sale at the barrier.

35.13 Explain why barriers cluster across issuers, why no issuer can observe the clustering, and

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what that implies about what should be monitored.

35.14 Explain why a note linked to the worst of three indices pays roughly double the coupon, and why that is a measure of risk rather than of value.

35.15 List the five evaluation steps of Section 35.11, state what each is designed to reveal, and apply them to Example 35.2's note in one sentence.

FURTHER QUESTIONS

35.16 Example 35.1 shows participation nearly doubling with the issuer's funding rate. Derive the participation rate as a function of funding, maturity, and volatility; determine which of the three a buyer can most easily verify; and design a comparison metric that would rank notes across issuers on a like-for-like basis.

35.17 Example 35.2 finds a fair coupon of 18.42% with a 16.45% probability of loss. Compute the distribution of the investor's return rather than its mean, state its skewness, and use it to argue either that the product is or is not suitable for a buyer whose stated objective is income with capital preservation.

35.18 Example 35.6 shows the hedge delta jumping by 0.19 at the barrier. Aggregate the flow: assume a market-wide notional and a barrier distribution, derive the total hedging demand as a function of the index level, and identify the level at which the withdrawal of the bid is largest.

35.19 Section 35.7 argues that a stabilizer which works until it stops is not a stabilizer. Assess the claim rigorously: construct the case that autocallable hedging reduces realized volatility on average, construct the case that it redistributes volatility into the tail, and state what data would settle which effect dominates.

35.20 Section 35.10 argues the margin is largest where the price is hardest to verify. Test this across the products in Chapters 33–35: rank them by verifiability and by typical margin, and state whether the relationship holds and what explains any exception.

35.21 Section 35.4 argues the exit is worst when most wanted. Quantify it: model the issuer's bid as fair value less a spread that widens with volatility, and compute the loss an investor takes by exiting after a 20% fall relative to holding to maturity.

35.22 Section 35.12 argues that a suitability test should be framed around payoff shape rather than stated risk tolerance. Design such a test: specify the statistics it would require, the thresholds it would apply, and how it would classify each product in this chapter. Then identify the product it misclassifies and say why.

35.23 The estimated-initial-value disclosure tells a buyer what the note is worth on the day it is issued. Determine what it does *not* tell them, propose one further disclosure that would close the largest remaining gap, and explain why an issuer would resist it.

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